

SMAYAN A. KHANNA

Cell: US: +1 (312) 278 5609
Chicago, Illinois, 60615

Email: smayan@uchicago.edu
GitHub: <https://github.com/SmayanKhanna>

EDUCATION

The University of Chicago, Chicago, IL

September 2025 – June 2026

- Master of Science (MS) in Computer Science
- Advanced Scholars Program (4+1)

The University of Chicago, Chicago, IL

September 2021 – June 2025

- Bachelor of Science (BS) in Physics with a Computer Science Minor
- **Cum Laude, Dean's List (21-22, 23-24 & 24-25), CGPA: 3.872/4, Major GPA: 3.92/4**

Relevant University Coursework: Machine Learning, NLP, Deep Active Learning, Algorithms Analysis, Probability, Data structures, Discrete Mathematics, Linear Algebra, Differential Equations, Quantum Mechanics, Numerical Methods

PAPERS & PUBLICATIONS

Jones, M., **Khanna, S.**, & Ferguson, A. (2024). FlowBack: Generalized Flow-Matching Approach for Biomolecular Backmapping (2024). Published to *Journal of Chemical Information and Modeling (JCIM)*, ACS Publications.

Jones, M., **Khanna, S.**, & Ferguson, A. (2024). FlowBack: A Flow-matching Approach for Generative Backmapping of Macromolecules. *ICML 2024 Workshop on Machine Learning for Life Sciences (ML4LMS)*. [Oral Presentation](#). Vienna, Austria.

Ashwood, B., Jones, M. S., Radakovic, A., **Khanna, S.**, Lee, Y., Sachleben, J. R., Szostak, J. W., Ferguson, A. L., & Tokmakoff, A. (2023). Thermodynamics and kinetics of DNA and RNA dinucleotide hybridization to gaps and overhangs. *Biophysical Journal*, 122(16). 3223–3339. <https://doi.org/10.1016/j.bpj.2023.07.009>.

PRESENTATIONS

FlowBack: A Flow-matching Approach for Generative Backmapping of Biomolecules. Poster presented at the *UChicago Undergraduate Research Symposium*, Chicago, IL. 2024.

Thermodynamics and kinetics of DNA and RNA dinucleotide hybridization to gaps and overhangs. Poster presented at the *Midstates Undergraduate Research Symposia*, Chicago, IL. 2023.

RESEARCH EXPERIENCE

University of Chicago - James Franck Institute, Vitelli Group, Chicago, IL

December 2024 – Present

Undergraduate Research Assistant

- Researching using graph neural networks to achieve optimal coarse-graining for inhomogeneous graph data
- Working on a first-author paper on evaluating the effectiveness of Graph Contrastive Learning (GCL) for unsupervised representation learning benchmarks for graph-classification

Massachusetts Institute of Technology (MIT), Dunkel Group, Cambridge, MA

June 2023 – August 2023

Undergraduate Research Intern

- Conducted research with Prof. Jörn Dunkel and the Flavell Lab on the mathematical brain-behavior link in *C. elegans*
- Developed a Julia pipeline which facilitated the analysis and visualization of neural trace activity datasets
- Applied PCA, graph theory, and compressed sensing in Julia to perform the dimensionality reduction of graph neural circuits
- Identified a superior neural basis for the signal reconstruction of timeseries data using Matching Pursuit, outperforming randomly sampled bases by 20%

Pritzker School of Molecular Engineering, Ferguson Lab, Chicago, IL

June 2022 – June 2024

Undergraduate Research Assistant (Sep 2022 – Jun 2024)

- Co-developed a generative model using Conditional Flow Matching and Equivariant Graph Neural Networks (EGNNs) to back-map protein and DNA traces from coarse-grained simulations
- Trained and fine-tuned the model on DNA-protein structures, optimizing inference time by evaluating different ODE solvers and achieving over 99% reconstruction accuracy, outperforming previous models by 3%

Undergraduate Research Intern (May 2022 – Aug 2022)

- Conducted research with the Tokmakoff Group and Prof. Andrew Ferguson on T-jump spectroscopy by generating all-atom molecular dynamics simulations of DNA
- Streamlined the creation of nucleotide topologies, replica exchange simulations, and cluster jobs through automation scripts

PROFESSIONAL EXPERIENCE

Motilal Oswal Financial Services (MOFSL), Mumbai, India

June 2024 – August 2024

Data-science Intern - Stock Prediction Model Development

- Implemented a pipeline using PCA and RFE for dimensionality reduction and feature-selection for stock prediction using machine learning
- Performed exploratory data analysis (EDA), outlier removal, and trained XGBoost models on NIFTY500 stocks
- Developed a back-testing framework, achieving over 60% prediction accuracy, and explored hyperparameter optimization using Random Search and Bayesian Search

PROJECTS

Deep Active Learning with Graph Neural Networks

April 2025 – June 2025

- Developed a Deep Active Learning (DAL) pipeline for scalable graph classification using Graph Neural Networks (GNNs) in Pytorch Geometric

Investigating Linguistic Complexity in LLMs

January 2025 – March 2025

- Developed a novel Activation Patching & Vector Patching method using Transformer Lens to probe Syntactic & Linguistic complexity in GPT-2
- Selected as Best Class Project (faculty-rated) for CMSC 25700: *Natural Language Processing*

Bot Checkers

January 2023 – May 2023

- Developed the GUI for an N-boards checkers game with integrated bot logic using Pygame

TEACHING EXPERIENCE

University of Chicago, Chicago, IL

March 2025 – Present

Teaching Assistant (TA)

- Sole TA for DATA 35900, a PhD level course on Differential Privacy & Algorithmic Fairness
- Assisted in course-organization, grading and assignment creation

The Science Biome, Mumbai, India

May 2016 - Present

Founder

- Researched, wrote, and animated 31 science videos on physics, math, and cosmology
- Amassed 2K subscribers and 90K+ views on the YouTube channel ([link](#))
- Recognized by Cornell Prof. Steven Strogatz for a video project on the applications of Differential Equations to HIV research and treatment ([tweet](#), [video](#))
- Interviewed by the Mumbai Mirror ([link](#)), and the national newspaper, The Times of India ([link](#))

AWARDS

2023-2024 Quad Undergraduate Research Scholarship, *University of Chicago*. \$5000.

2023 Jeff Metcalf Internship Fellowship. *University of Chicago*. \$4000.

2021 Runner-up, Strogatz Math Prize. *National Museum of Mathematics (MoMath), USA*. \$314.159.

2020 Breakthrough Junior Challenge Semifinalist, *Breakthrough Prize Organization, Khan Academy & National Geographic*.

PROGRAMMING SKILLS

Python: Pytorch, Pytorch Geometric, HuggingFace Transformers, Scikit Learn, Pandas, TensorFlow, Numpy

Julia: DifferentialEquations.jl, MultivariateStats.jl, Graphs.jl

Tools: Git, CUDA, SLURM